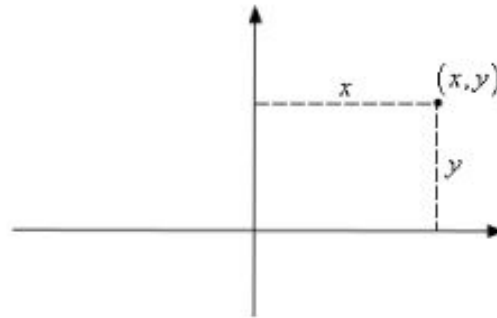


Coordinate Geometry

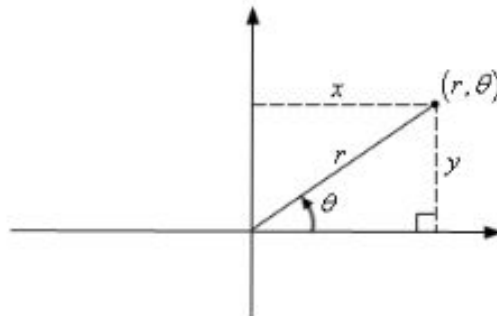
Polar Coordinate

Up to this point we've dealt exclusively with the Cartesian (or Rectangular, or x - y) coordinate system. However, as we will see, this is not always the easiest coordinate system to work in. So, in this section we will start looking at the polar coordinate system.

Coordinate systems are really nothing more than a way to define a point in space. For instance in the Cartesian coordinate system a point is given the coordinates (x, y) and we use this to define the point by starting at the origin and then moving x units horizontally followed by y units vertically. This is shown in the sketch below.



This is not, however, the only way to define a point in two dimensional space. Instead of moving vertically and horizontally from the origin to get to the point we could instead go straight out of the origin until we hit the point and then determine the angle this line makes with the positive x -axis. We could then use the distance of the point from the origin and the amount we needed to rotate from the positive x -axis as the coordinates of the point. This is shown in the sketch below.



Coordinates in this form are called **polar coordinates**.

Coordinate conversions

Polar to Cartesian Conversion Formulas

$$x = r \cos \theta \qquad y = r \sin \theta$$

Converting from Cartesian is almost as easy. Let's first notice the following.

$$\begin{aligned} x^2 + y^2 &= (r \cos \theta)^2 + (r \sin \theta)^2 \\ &= r^2 \cos^2 \theta + r^2 \sin^2 \theta \\ &= r^2 (\cos^2 \theta + \sin^2 \theta) = r^2 \end{aligned}$$

Cartesian to Polar Conversion Formulas

$$\begin{aligned} r^2 &= x^2 + y^2 & r &= \sqrt{x^2 + y^2} \\ \theta_1 &= \tan^{-1} \left(\frac{y}{x} \right) & \text{OR} & \theta_2 = \theta_1 + \pi \end{aligned}$$

(a) Convert $\left(-4, \frac{2\pi}{3}\right)$ into Cartesian coordinates.

This conversion is easy enough. All we need to do is plug the points into the formulas.

$$x = -4 \cos\left(\frac{2\pi}{3}\right) = -4 \left(-\frac{1}{2}\right) = 2$$

$$y = -4 \sin\left(\frac{2\pi}{3}\right) = -4 \left(\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$$

So, in Cartesian coordinates this point is $(2, -2\sqrt{3})$.

(b) Convert $(-1, -1)$ into polar coordinates.

Let's first get r .

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

Now, let's get θ .

$$\theta = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

This is not the correct angle however. This value of θ is in the first quadrant and the point we've been given is in the third quadrant. As noted above we can get the correct angle by adding π onto this. Therefore, the actual angle is,

$$\theta = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$$

So, in polar coordinates the point is $\left(\sqrt{2}, \frac{5\pi}{4}\right)$. Note as well that we could have used the first θ that we got by using a negative r . In this case the point could also be written in polar coordinates as $\left(-\sqrt{2}, \frac{\pi}{4}\right)$.

Coordinate Conversion Example

Equation Conversion Examples

Example 2 Convert each of the following into an equation in the given coordinate system.

(a) Convert $2x - 5x^3 = 1 + xy$ into polar coordinates.

(b) Convert $r = -8 \cos \theta$ into Cartesian coordinates.

(a) Convert $2x - 5x^3 = 1 + xy$ into polar coordinates.

In this case there really isn't much to do other than plugging in the formulas for x and y (i.e. the Cartesian coordinates) in terms of r and θ (i.e. the polar coordinates).

$$\begin{aligned} 2(r \cos \theta) - 5(r \cos \theta)^3 &= 1 + (r \cos \theta)(r \sin \theta) \\ 2r \cos \theta - 5r^3 \cos^3 \theta &= 1 + r^2 \cos \theta \sin \theta \end{aligned}$$

(b) Convert $r = -8 \cos \theta$ into Cartesian coordinates.

This one is a little trickier, but not by much. First notice that we could substitute straight for the r . However, there is no straight substitution for the cosine that will give us only Cartesian coordinates. If we had an r on the right along with the cosine then we could do a direct substitution. So, if an r on the right side would be convenient let's put one there, just don't forget to put one on the left side as well.

$$r^2 = -8r \cos \theta$$

We can now make some substitutions that will convert this into Cartesian coordinates.

$$x^2 + y^2 = -8x$$

Circles in Polar coordinates

Circles

Let's take a look at the equations of circles in polar coordinates.

1. $r = a$.

This equation is saying that no matter what angle we've got the distance from the origin must be a . If you think about it that is exactly the definition of a circle of radius a centered at the origin.

So, this is a circle of radius a centered at the origin. This is also one of the reasons why we might want to work in polar coordinates. The equation of a circle centered at the origin has a very nice equation, unlike the corresponding equation in Cartesian coordinates.

2. $r = 2a \cos \theta$.

We looked at a specific example of one of these when we were converting equations to Cartesian coordinates.

This is a circle of radius $|a|$ and center $(a, 0)$. Note that a might be negative (as it was in our example above) and so the absolute value bars are required on the radius. They should not be used however on the center.

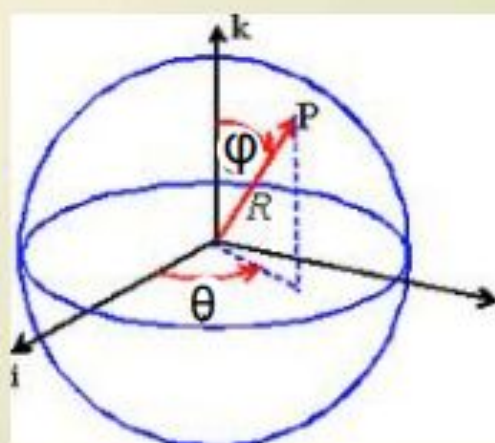
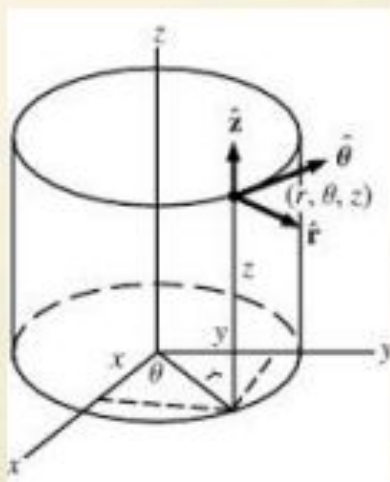
3. $r = 2b \sin \theta$.

This is similar to the previous one. It is a circle of radius $|b|$ and center $(0, b)$.

4. $r = 2a \cos \theta + 2b \sin \theta$.

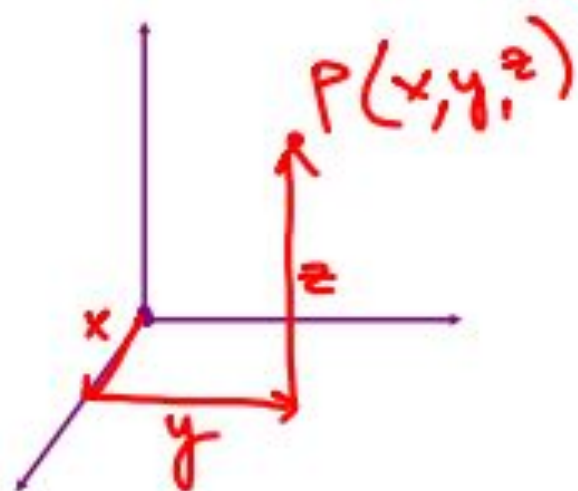
This is a combination of the previous two and by completing the square twice it can be shown that this is a circle of radius $\sqrt{a^2 + b^2}$ and center (a, b) . In other words, this is the general equation of a circle that isn't centered at the origin.

Cylindrical and Spherical Coordinates

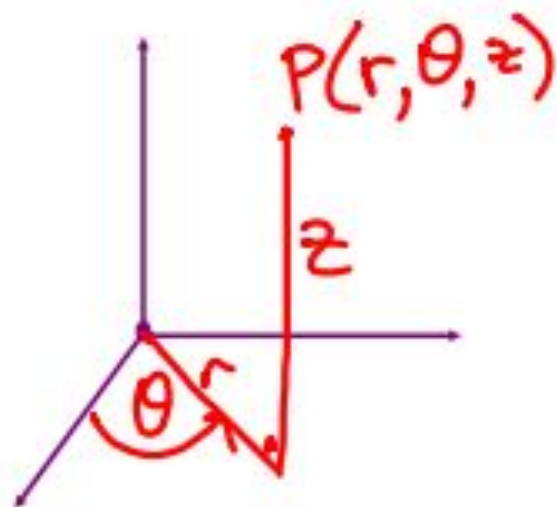


We can describe a point, P , in three different ways.

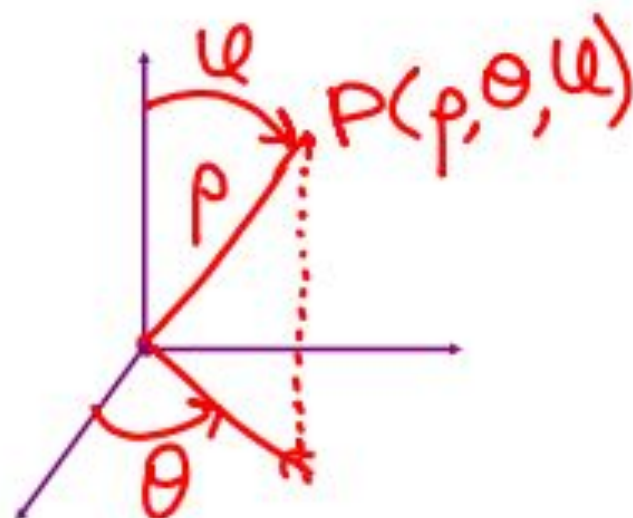
Cartesian



Cylindrical



Spherical



Cylindrical coordinate system

In the cylindrical coordinate system, a point in space (Figure 12.7.1) is represented by the ordered triple (r, θ, z) , where

- (r, θ) are the polar coordinates of the point's projection in the xy -plane
- z is the usual z -**coordinate** in the Cartesian coordinate system

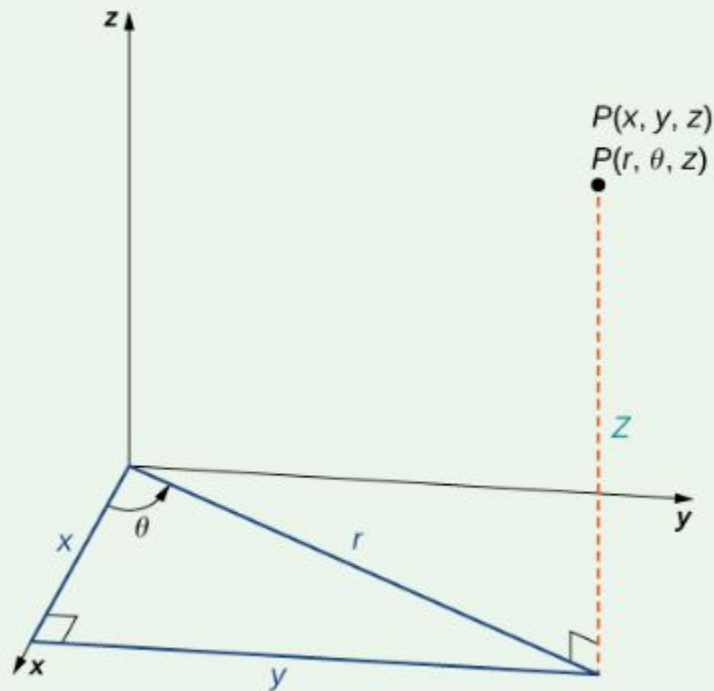


Figure 12.7.1: The right triangle lies in the xy -plane. The length of the hypotenuse is r and θ is the measure of the angle formed by the positive x -axis and the hypotenuse. The z -coordinate describes the location of the point above or below the xy -plane.

Equations of conversion

The rectangular coordinates (x, y, z) and the cylindrical coordinates (r, θ, z) of a point are related as follows:

These equations are used to convert from cylindrical coordinates to rectangular coordinates.

- $x = r \cos \theta$
- $y = r \sin \theta$
- $z = z$

These equations are used to convert from rectangular coordinates to cylindrical coordinates

1. $r^2 = x^2 + y^2$
2. $\tan \theta = \frac{y}{x}$
3. $z = z$

Spherical coordinate system

In the *spherical coordinate system*, a point P in space (Figure 12.7.9) is represented by the ordered triple (ρ, θ, φ) where

- ρ (the Greek letter rho) is the distance between P and the origin ($\rho \neq 0$);
- θ is the same angle used to describe the location in cylindrical coordinates;
- φ (the Greek letter phi) is the angle formed by the positive z -axis and line segment $\overline{OP}$, where O is the origin and $0 \leq \varphi \leq \pi$.

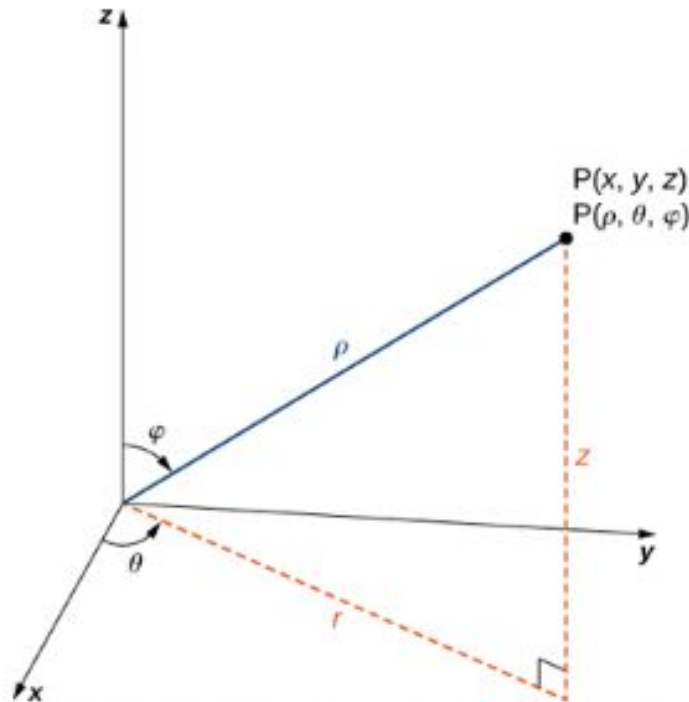


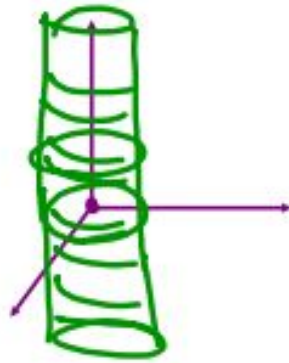
Figure 12.7.9: The relationship among spherical, rectangular, and cylindrical coordinates.

By convention, the origin is represented as $(0, 0, 0)$ in spherical coordinates.

Cylindrical coordinate surfaces

Easy Surfaces in Cylindrical Coordinates

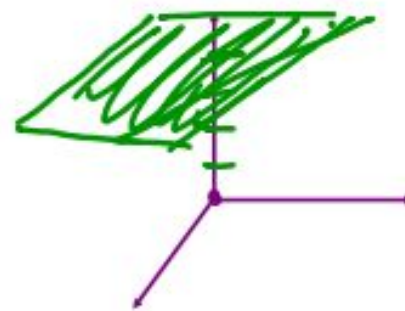
a) $r = 1$



b) $\theta = \pi/3$

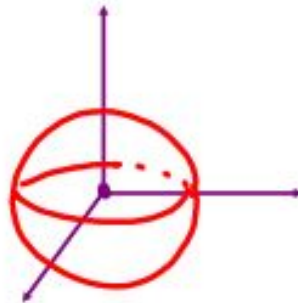


c) $z = 4$

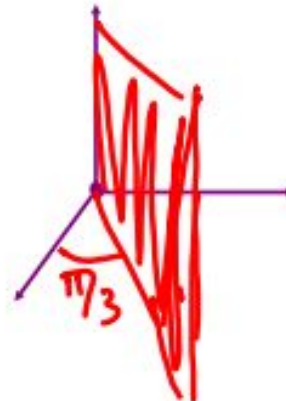


Easy Surfaces in Spherical Coordinates

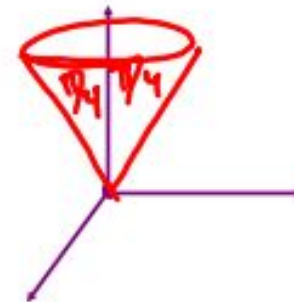
a) $\rho = 1$



b) $\theta = \pi/3$



c) $\varphi = \pi/4$



Cylindrical to rectangular conversion

Plot the point with cylindrical coordinates $(4, \frac{2\pi}{3}, -2)$ and express its location in rectangular coordinates.

Solution

Conversion from cylindrical to rectangular coordinates requires a simple application of the equations listed in Conversion between Cylindrical and Cartesian Coordinates:

$$x = r \cos \theta = 4 \cos \frac{2\pi}{3} = -2$$

$$y = r \sin \theta = 4 \sin \frac{2\pi}{3} = 2\sqrt{3}$$

$$z = -2$$

The point with cylindrical coordinates $(4, \frac{2\pi}{3}, -2)$ has rectangular coordinates $(-2, 2\sqrt{3}, -2)$ (Figure 12.7.5).

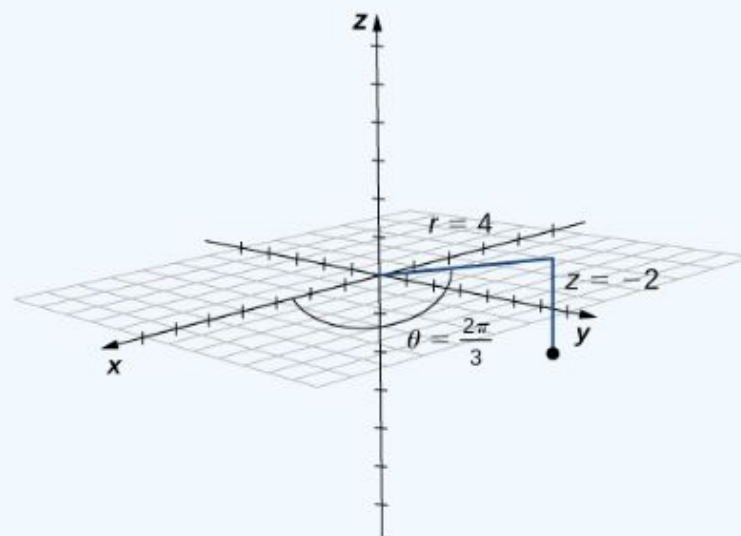


Figure 12.7.5: The projection of the point in the xy -plane is 4 units from the origin. The line from the origin to the point's projection forms an angle of $\frac{2\pi}{3}$ with the positive x -axis. The point lies 2 units below the xy -plane.

Rectangular to cylindrical conversion

Convert the rectangular coordinates $(1, -3, 5)$ to cylindrical coordinates.

Solution

Use the second set of equations from Conversion between Cylindrical and Cartesian Coordinates to translate from rectangular to cylindrical coordinates:

$$\begin{aligned}r^2 &= x^2 + y^2 \\r &= \pm\sqrt{1^2 + (-3)^2} \\&= \pm\sqrt{10}.\end{aligned}$$

We choose the positive square root, so $r = \sqrt{10}$. Now, we apply the formula to find θ . In this case, y is negative and x is positive, which means we must select the value of θ between $\frac{3\pi}{2}$ and 2π :

$$\begin{aligned}\tan \theta &= \frac{y}{x} &= \frac{-3}{1} \\ \theta &= \arctan(-3) &\approx 5.03 \text{ rad.}\end{aligned}$$

In this case, the z -coordinates are the same in both rectangular and cylindrical coordinates:

$$z = 5.$$

The point with rectangular coordinates $(1, -3, 5)$ has cylindrical coordinates approximately equal to $(\sqrt{10}, 5.03, 5)$.

Rectangular to spherical and cylindrical conversion

Convert the rectangular coordinates $(-1, 1, \sqrt{6})$ to both spherical and cylindrical coordinates.

Solution

Start by converting from rectangular to spherical coordinates:

$$\begin{aligned}\rho^2 &= x^2 + y^2 + z^2 = (-1)^2 + 1^2 + (\sqrt{6})^2 = 8 \\ \tan \theta &= \frac{1}{-1} \\ \rho &= 2\sqrt{2} \text{ and } \theta = \arctan(-1) = \frac{3\pi}{4}.\end{aligned}$$

Because $(x, y) = (-1, 1)$, then the correct choice for θ is $\frac{3\pi}{4}$.

There are actually two ways to identify φ . We can use the equation $\varphi = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$. A more simple approach, however, is to use equation $z = \rho \cos \varphi$. We know that $z = \sqrt{6}$ and $\rho = 2\sqrt{2}$, so

$$\sqrt{6} = 2\sqrt{2} \cos \varphi, \text{ so } \cos \varphi = \frac{\sqrt{6}}{2\sqrt{2}} = \frac{\sqrt{3}}{2}$$

and therefore $\varphi = \frac{\pi}{6}$. The spherical coordinates of the point are $(2\sqrt{2}, \frac{3\pi}{4}, \frac{\pi}{6})$.

To find the cylindrical coordinates for the point, we need only find r :

$$r = \rho \sin \varphi = 2\sqrt{2} \sin\left(\frac{\pi}{6}\right) = \sqrt{2}.$$

The cylindrical coordinates for the point are $(\sqrt{2}, \frac{3\pi}{4}, \sqrt{6})$.

Coordinate Conversion Practice Problems

Cylindrical Coordinates

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \left\{ \begin{array}{l} \text{from} \\ \text{cyl. to} \\ \text{Cartesian} \end{array} \right.$$

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \tan \theta = y/x \\ z = z \end{cases} \left\{ \begin{array}{l} \text{from Cartesian} \\ \text{to cylindrical} \end{array} \right.$$

$$r \geq 0, \theta \in [0, 2\pi)$$

$$\text{note: } \tan \theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Spherical Coordinates

$$x = \rho \sin \varphi \cos \theta$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

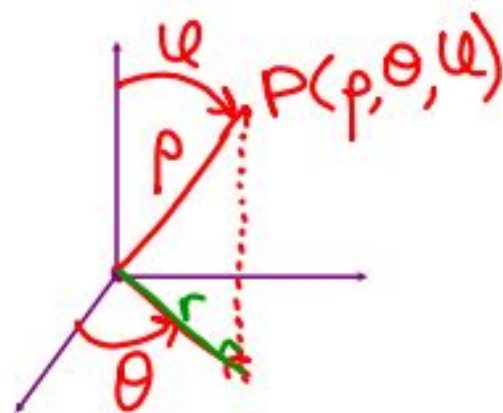
$$y = \rho \sin \varphi \sin \theta$$

$$\tan \theta = y/x$$

$$z = \rho \cos \varphi$$

$$\cos \varphi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

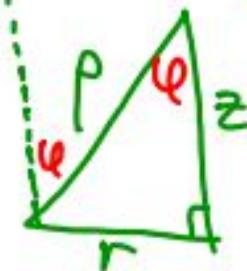
$$\rho \geq 0, \theta \in [0, 2\pi), \text{ and } \varphi \in [0, \pi]$$



pull out right Δ :

$$r = \rho \sin \varphi$$

$$z = \rho \cos \varphi$$



$$\sin \varphi = \frac{r}{\rho}$$

$$\cos \varphi = \frac{z}{\rho}$$

EX 1 Convert the coordinates as indicated

a) $(3, \pi/3, -4)$ from cylindrical to Cartesian.

$$x = r \cos \theta = 3 \cos(\pi/3) = 3 \left(\frac{1}{2}\right) = \frac{3}{2}$$

$$y = r \sin \theta = 3 \sin(\pi/3) = 3 \left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}$$

$$z = z = -4$$

$$\boxed{\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}, -4\right)}$$

b) $(-2, 2, 3)$ from Cartesian to cylindrical.

in xy-plane
•
+

$$r = \sqrt{x^2 + y^2} = \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2}$$

$$\tan \theta = \frac{y}{x} \Leftrightarrow \tan \theta = \frac{2}{-2} = -1$$

$$z = z = 3$$

$$\theta = 3\pi/4$$

$$\boxed{\left(2\sqrt{2}, \frac{3\pi}{4}, 3\right)}$$

EX 2 Convert the coordinates as indicated

a) $(8, \pi/4, \pi/6)$ from spherical to Cartesian.

$$x = \rho \cos \theta \sin \varphi = 8 \cos(\pi/4) \sin(\pi/6) = 8 \left(\frac{\sqrt{2}}{2}\right) \left(\frac{1}{2}\right)$$

$$y = \rho \sin \theta \sin \varphi = 8 \sin(\pi/4) \sin(\pi/6)$$

$$= 2\sqrt{2}$$

$$z = \rho \cos \varphi$$

$$= 2\sqrt{2}$$

$$= 8 \cos(\pi/6) = 8 \left(\frac{\sqrt{3}}{2}\right) = 4\sqrt{3}$$

$$(2\sqrt{2}, 2\sqrt{2}, 4\sqrt{3})$$

b) $(2\sqrt{3}, 6, -4)$ from Cartesian to spherical.

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{(2\sqrt{3})^2 + 6^2 + 4^2} = \sqrt{12 + 36 + 16} = \sqrt{64} = 8$$

$$\tan \theta = \frac{y}{x}$$

$$\cos \varphi = \frac{z}{\rho}$$

$$\cos \varphi = \frac{-4}{8} = -\frac{1}{2}$$

$$\varphi \in [0, \pi] \quad \varphi = 2\pi/3$$

$$(8, \pi/3, 2\pi/3)$$

$$\tan \theta = \frac{6}{2\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\tan \theta = \frac{\sqrt{3}/2}{1/2}$$

$$\theta = \pi/3$$

in xy-plane:
• (in Q1)


EX 3 Convert from cylindrical to spherical coordinates.

$$\begin{matrix} r & \theta & z \\ (1, \pi/2, 1) \end{matrix}$$

$$\theta = \pi/2$$

$$\theta = \theta$$

$$r = \rho \sin \varphi$$

$$z = \rho \cos \varphi$$

$$\left. \begin{matrix} r = \rho \sin \varphi \\ z = \rho \cos \varphi \end{matrix} \right\} \Rightarrow \begin{matrix} \textcircled{1} \frac{r}{z} = \tan \varphi \\ \textcircled{2} \rho = \frac{z}{\cos \varphi} \end{matrix}$$

$$\textcircled{1} \frac{1}{1} = \tan \varphi \quad (\varphi \in [0, \pi])$$

$$\varphi = \pi/4$$

$$\textcircled{2} \rho = \frac{1}{\cos(\pi/4)} = \frac{1}{\sqrt{2}/2} = \sqrt{2}$$

$$\boxed{(\sqrt{2}, \pi/2, \pi/4)}$$

EX 4 Make the required change in the given equation.

a) $x^2 - y^2 = 25$ to cylindrical coordinates.

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ r^2 \cos^2 \theta - r^2 \sin^2 \theta &= 25 \\ r^2 (\cos^2 \theta - \sin^2 \theta) &= 25 \\ r^2 \cos(2\theta) &= 25 \end{aligned} \rightarrow r^2 = 25 \sec(2\theta)$$

b) $x^2 + y^2 - z^2 = 1$ to spherical coordinates.

$$\begin{aligned} x &= \rho \cos \theta \sin \varphi \\ y &= \rho \sin \theta \sin \varphi \\ z &= \rho \cos \varphi \end{aligned} \left| \begin{aligned} \rho^2 \cos^2 \theta \sin^2 \varphi + \rho^2 \sin^2 \theta \sin^2 \varphi - \rho^2 \cos^2 \varphi &= 1 \\ \rho^2 \sin^2 \varphi (\cos^2 \theta + \sin^2 \theta) - \rho^2 \cos^2 \varphi &= 1 \\ \rho^2 (\sin^2 \varphi - \cos^2 \varphi) &= 1 \\ \rho^2 (-\cos(2\varphi)) &= 1 \end{aligned} \right.$$

c) $\rho = 2 \cos \varphi$ to cylindrical coordinates.

$$\Rightarrow \boxed{\rho^2 = -\sec(2\varphi)}$$

$$\begin{aligned} r &= \rho \sin \varphi \\ z &= \rho \cos \varphi \end{aligned} \left| \begin{aligned} \rho^2 &= 2\rho \cos \varphi \\ \rho^2 &= 2z \end{aligned} \right. \\ r^2 + z^2 = \rho^2 &\Rightarrow r^2 + z^2 = 2z \\ r^2 &= 2z - z^2 \end{aligned}$$

EX 4 Make the required change in the given equation (continued).

d) $x + y + z = 1$ to spherical coordinates.

$$\rho \sin \varphi \cos \theta + \rho \sin \varphi \sin \theta + \rho \cos \varphi = 1$$

e) $r = 2 \sin \theta$ to Cartesian coordinates.

$$r^2 = 2r \sin \theta$$

$$x^2 + y^2 = 2y$$

(cylinder)

$$x^2 + y^2 - 2y + 1 = 0 + 1$$

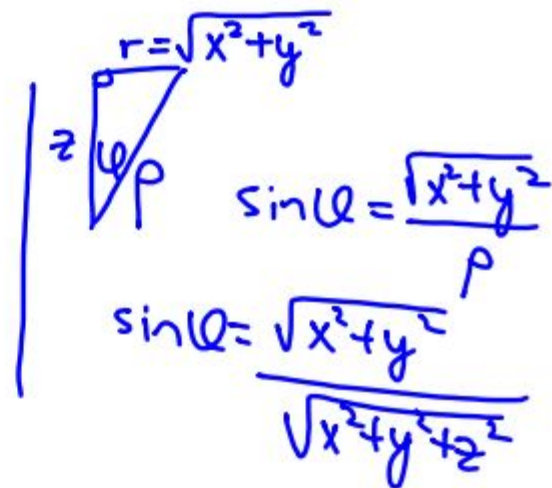
$$x^2 + (y-1)^2 = 1$$

f) $\rho \sin \theta = 1$ to Cartesian coordinates.

$$\rho \sin \varphi \sin \theta = \sin \varphi$$

$$y = \sin \varphi$$

$$y = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$


$$\sin \varphi = \frac{\sqrt{x^2 + y^2}}{\rho}$$
$$\sin \varphi = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

**PRACTICE SUMS FROM HOWARD ANTON 10th
Edition Page-837, Exercise: 11.8**

Express the equations in cylindrical coordinates as equations in rectangular coordinates

$$\begin{aligned}\underline{19.} \quad \rho &= 3 \\ \therefore \rho^2 &= x^2 + y^2 \\ \therefore 3^2 &= x^2 + y^2 \\ \Rightarrow x^2 + y^2 &= 9\end{aligned}$$

$$\begin{aligned}\underline{23.} \quad \rho &= 4 \sin \theta \\ \text{we know, } y &= \rho \sin \theta. \\ \therefore \text{If we make } \sin \theta \text{ subject:} \\ \sin \theta &= \frac{\rho}{4}, \quad \sin \theta = \frac{y}{\rho} \\ \therefore \frac{\rho}{4} &= \frac{y}{\rho} \\ \therefore 4y &= \rho^2 \\ \text{Now, } \rho^2 &= x^2 + y^2 \\ \therefore 4y &= x^2 + y^2\end{aligned}$$

$$\begin{aligned}\underline{25.} \quad \rho^2 + z^2 &= 1 \\ \text{we know, } \rho^2 &= x^2 + y^2 \\ \therefore x^2 + y^2 + z^2 &= 1\end{aligned}$$

$$\begin{aligned}\underline{26.} \quad \rho^2 \cos 2\theta &= z \\ \text{we know, } \cos 2\theta &= 2\cos^2 \theta - 1 \\ \therefore \rho^2 (2\cos^2 \theta - 1) &= z \\ \Rightarrow 2\rho^2 \cos^2 \theta - \rho^2 &= z \\ \therefore \text{We know, } x &= \rho \cos \theta \\ \therefore x^2 &= \rho^2 \cos^2 \theta \\ \therefore 2x^2 - (x^2 + y^2) &= z \\ \Rightarrow 2x^2 - x^2 - y^2 &= z \\ \Rightarrow x^2 - y^2 &= z\end{aligned}$$

Express the equations in spherical coordinates as equations in rectangular coordinates

$$\begin{aligned} 27. \quad \rho &= 3 \\ \text{we know, } \rho &= \sqrt{x^2 + y^2 + z^2} \\ \therefore \sqrt{x^2 + y^2 + z^2} &= 3 \\ \therefore x^2 + y^2 + z^2 &= 9 \end{aligned}$$

$$\begin{aligned} 31. \quad \rho &= 4 \cos \phi \\ \therefore \cos \phi &= \frac{\rho}{4} \\ \text{we know, } z &= \rho \cos \phi \\ \therefore \cos \phi &= \frac{z}{\rho} \end{aligned}$$

$$\begin{aligned} \therefore \frac{\rho}{4} &= \frac{z}{\rho} \\ \therefore \rho^2 &= 4z \end{aligned}$$

$$\begin{aligned} \text{we know, } \rho &= \sqrt{x^2 + y^2 + z^2} \\ \therefore \rho^2 &= x^2 + y^2 + z^2 \\ \text{putting the value of } \rho^2 \\ \therefore 4z &= x^2 + y^2 + z^2 \end{aligned}$$

Express the equations in rectangular coordinates as equations in (a) Cylindrical coordinates (b) Spherical coordinates

$$35. \quad z = 3$$

a) $z = 3$

b) $\rho \cos \phi = 3$

$$38. \quad z = \sqrt{3x^2 + 3y^2}$$

a) $\therefore z = \sqrt{3(x^2 + y^2)}$
 $\Rightarrow z = \sqrt{3\rho^2}$
 $\Rightarrow z = \sqrt{3} \rho$

b) We know, $z = \rho \cos \phi$
putting the value of z
 $\sqrt{3} \rho = \rho \cos \phi$
 $\Rightarrow \sqrt{3} \times \rho \sin \phi = \rho \cos \phi$
 $\Rightarrow \frac{\rho \sin \phi}{\rho \cos \phi} = \frac{1}{\sqrt{3}}$
 $\Rightarrow \tan \phi = \frac{1}{\sqrt{3}}$
 $\Rightarrow \phi = \pi/6$

We know, $y = \rho \sin \theta$
 $\Rightarrow \frac{y}{\sin \theta} = \rho$

Again, $y = \rho \sin \phi \sin \theta$
 $\Rightarrow \rho \sin \phi = \frac{y}{\sin \theta} = \rho$
 $\therefore \rho = \rho \sin \phi$

Express the equations in rectangular coordinates as equations in (a) Cylindrical coordinates (b) Spherical coordinates

41. $x^2 + y^2 + z^2 = 9$

a) $\rho^2 + z^2 = 9$

b) $\rho = \sqrt{x^2 + y^2 + z^2}$

$\therefore \rho = \sqrt{9}$

$\rho = 3$

45. $x^2 = 16 - z^2$

a) $(\rho \cos \theta)^2 = 16 - z^2$

$\Rightarrow \rho^2 \cos^2 \theta = 16 - z^2$

b) $x^2 = 16 - z^2$

$\Rightarrow (\rho \sin \phi \cos \theta)^2 = 16 - (\rho \cos \phi)^2$

$\Rightarrow \rho^2 \sin^2 \phi \cos^2 \theta = 16 - \rho^2 \cos^2 \phi$

Spherical Equation Surface Sums

Describe the surfaces with the given spherical equations.

a. $\theta = \frac{\pi}{3}$

b. $\varphi = \frac{5\pi}{6}$

c. $\rho = 6$

d. $\rho = \sin \theta \sin \varphi$

Solution

a. The variable θ represents the measure of the same angle in both the cylindrical and spherical coordinate systems. Points with coordinates $(\rho, \frac{\pi}{3}, \varphi)$ lie on the plane that forms angle $\theta = \frac{\pi}{3}$ with the positive x -axis. Because $\rho > 0$, the surface described by equation $\theta = \frac{\pi}{3}$ is the half-plane shown in Figure 12.7.13.

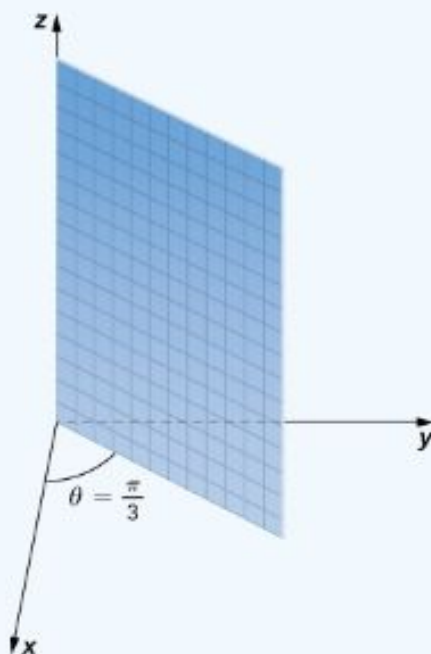


Figure 12.7.13: The surface described by equation $\theta = \frac{\pi}{3}$ is a half-plane.

b. Equation $\varphi = \frac{5\pi}{6}$ describes all points in the spherical coordinate system that lie on a line from the origin forming an angle measuring $\frac{5\pi}{6}$ rad with the positive z -axis. These points form a half-cone (Figure 12.7.14). Because there is only one value for φ that is measured from the positive z -axis, we do not get the full cone (with two pieces).

b. Equation $\varphi = \frac{5\pi}{6}$ describes all points in the spherical coordinate system that lie on a line from the origin forming an angle measuring $\frac{5\pi}{6}$ rad with the positive z -axis. These points form a half-cone (Figure 12.7.14). Because there is only one value for φ that is measured from the positive z -axis, we do not get the full cone (with two pieces).

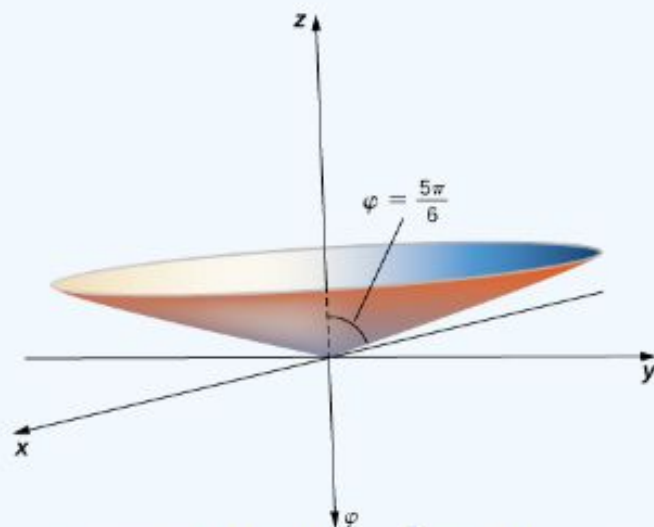


Figure 12.7.14: The equation $\varphi = \frac{5\pi}{6}$ describes a cone.

To find the equation in rectangular coordinates, use equation $\varphi = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$.

$$\frac{5\pi}{6} = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$$

$$\cos \frac{5\pi}{6} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$-\frac{\sqrt{3}}{2} = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\frac{3}{4} = \frac{z^2}{x^2 + y^2 + z^2}$$

$$\frac{3x^2}{4} + \frac{3y^2}{4} + \frac{3z^2}{4} = z^2$$

$$\frac{3x^2}{4} + \frac{3y^2}{4} - \frac{z^2}{4} = 0.$$

This is the equation of a cone centered on the z -axis.

c. Equation $\rho = 6$ describes the set of all points 6 units away from the origin—a sphere with radius 6 (Figure 12.7.15).

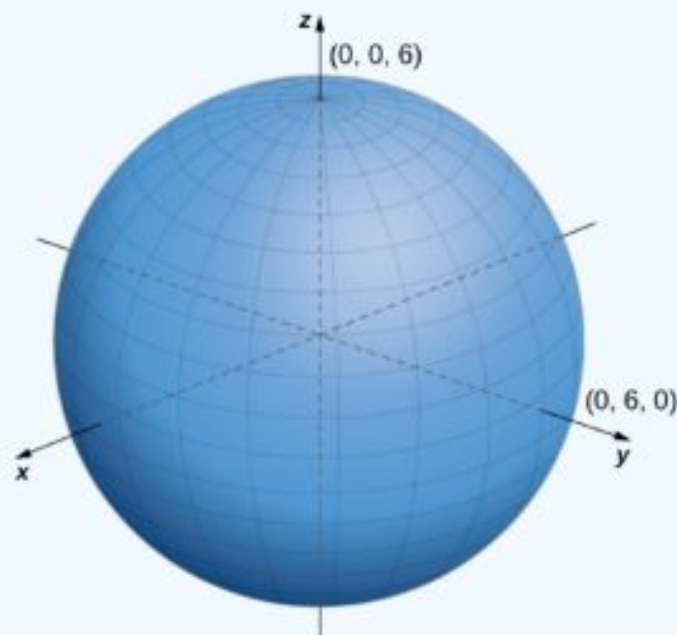


Figure 12.7.15: Equation $\rho = 6$ describes a sphere with radius 6.

d. To identify this surface, convert the equation from spherical to rectangular coordinates, using equations $y = \rho \sin \varphi \sin \theta$ and $\rho^2 = x^2 + y^2 + z^2$:

$$\rho = \sin \theta \sin \varphi$$

$\rho^2 = \rho \sin \theta \sin \varphi$ Multiply both sides of the equation by ρ .

$x^2 + y^2 + z^2 = y$ Substitute rectangular variables using the equations above.

$x^2 + y^2 - y + z^2 = 0$ Subtract y from both sides of the equation.

$x^2 + y^2 - y + \frac{1}{4} + z^2 = \frac{1}{4}$ Complete the square.

$x^2 + \left(y - \frac{1}{2}\right)^2 + z^2 = \frac{1}{4}$ Rewrite the middle terms as a perfect square.

The equation describes a sphere centered at point $(0, \frac{1}{2}, 0)$ with radius $\frac{1}{2}$.

Convert from spherical coordinates to rectangular coordinates

These equations are used to convert from spherical coordinates to rectangular coordinates.

- $x = \rho \sin \varphi \cos \theta$
- $y = \rho \sin \varphi \sin \theta$
- $z = \rho \cos \varphi$

Convert from rectangular coordinates to spherical coordinates

These equations are used to convert from rectangular coordinates to spherical coordinates.

- $\rho^2 = x^2 + y^2 + z^2$
- $\tan \theta = \frac{y}{x}$
- $\varphi = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right).$

Convert from spherical coordinates to cylindrical coordinates

These equations are used to convert from spherical coordinates to cylindrical coordinates.

- $r = \rho \sin \varphi$
- $\theta = \theta$
- $z = \rho \cos \varphi$

Convert from cylindrical coordinates to spherical coordinates

These equations are used to convert from cylindrical coordinates to spherical coordinates.

- $\rho = \sqrt{r^2 + z^2}$
- $\theta = \theta$
- $\varphi = \arccos\left(\frac{z}{\sqrt{r^2 + z^2}}\right)$

Summary